

**23** A police car has a two-tone siren emitting sound of frequencies of 700 Hz and 1000 Hz.

The police car is travelling at a speed of  $40.0 \text{ m s}^{-1}$  towards a stationary observer. The speed of sound in the air is  $340 \text{ m s}^{-1}$ .

What is the difference between the two frequencies of the sound that is heard by the observer?

**A** 268 Hz

**B** 300 Hz

**C** 335 Hz

**D** 340 Hz